Gate 3: Final scientific and submission audit

CONDITIONAL GO | Independent sign-off | 17 October 2026

1. Executive scientific verdict

CONDITIONAL GO for a controlled accounting-and-network-model paper. No known invalidating mathematical contradiction remains in the scoped claims. The three independently retained annual lower bounds and original B03 witnesses give strict guarded numerical separations; the adverse rolling and exposure controls remain visible. This decision is not journal acceptance, author approval, a secure-dispatch recommendation, or an interval certificate.

Required human release actions: approve author/affiliation and AI declarations; choose the technical-report license; separately authorize its Zenodo deposit; verify the actual report DOI; update the normal pre-deposit citation and regenerate a new versioned publication lock. Some upstream public/generator revisions remain unbound. Retain the stated numerical and engineering limits.

Scope used: exactly three annual M1-LP LP solves, no new M2 solve, no Stage II verification optimization, no additional debit family, no AC run and no frozen stage runner. Forty-one original saved objects and fourteen carried ledgers were freshly rechecked. All new scientific artifacts reside outside the protected repository.

2. Authoritative manuscript versions and hashes

The three explicit latest attachments are byte-identical to TPWRS_PaperB_AuditIntegrated_Complete_20261017.zip. Selection used the required corrected rounding, hybrid projection, guarded witness distinction, continuous-versus-generic M1 route, conservative premium controls, actuator diagnostics, and text-only AC validation. Older Final/specification and earlier submission packages were not selected.

Role	Authoritative SHA-256
main.tex	e21072f9995c6ddebf071c8cf57b62cced72c6e8faf37bdbc85312deddff3be0
technical_companion.tex	6eedfec3e9fc9934bde403efe16f5e3e7acba8c90e52c79ed2ab3a2eece4acbb
references.bib	ceb17e4b3a9816e107ec3f4d98f46b42894364a36628a8914e139f904f341651

AUTHORITY.json records exact attachment and ZIP paths/hashes. Revised copies are separate from all authoritative inputs. Edit logs distinguish new independent endpoints, outward display rounding, implementation-exposition corrections, bibliographic metadata, and editorial shortening. The source/report figure asset is copied from the selected latest ZIP without modifying its bytes.

3. Gate 1 outcomes and numerical meaning

Year	New guarded L	Original B03 U	δ	Margin	Reduction %
2029	57909827.749258	48058421.890843	0.057909828	9851405.800505	17.011630
2030	37782287.250540	32237842.052070	0.037782287	5544445.160687	14.674721
2031	23658821.383772	18740819.467405	0.023658821	4918001.892709	20.787180

Every annual physical/time/position identity and historical screening hash reconciles. The new masters use all positive-box actions and all 8760 hourly epigraphs, C0=0, and archived globally valid supports. This valid full-box relaxation avoids relying on historical coordinate-screening exactness. The source-compatible slopes match exactly; intercept summations differ only at roughly10^-11.

Gurobi 12.0.3, one thread and dual simplex reached OPTIMAL in 120.11,72.80 and 455.40 seconds, with full primal/dual vectors and bases saved. These new runtime/settings records do not replace historical two-stage policy timings. Presolve 2 and 10^-9 tolerances are explicitly different from the historical independent source profile.

Signs are projected and each actual binary-vector hourly multiplier sum is checked by exact rational arithmetic. The finite-box bound is computed by support-column and original physical-row aggregation paths; their discrepancies are0,7.45×10^-9 and 0. Downward guards are9.42×10^-5,2.93×10^-5 and 9.68×10^-6 full

residual units. Original witness coordinates pass exact box/FDE arithmetic. Independent upper replays have error estimates0.00605,0.00406 and 0.00235; deducting those additionally leaves strict margins.

These are CONDITIONAL PASS numerical certificates, with quantified standard floating-point analysis and complete reproducible vectors. They are not fully outward-rounded interval proofs. New lower endpoints are less than 0.003 full units below historical scalars; the displayed17.0%, 14.7%, 20.8% conclusions survive. Missing historical full vectors remain missing and are not retrospectively fabricated.

4. Final mathematics audit: proofs and counterexamples

Projection: when0≤ell≤b, the least relaxed activation is u=x/b on b>0. It satisfies the lower link and minimizes a debit nondecreasing in u. Substitution gives α/b+(1−α)/P, exactly R_(1−α); b=0 is fixed to zero. The isolated convex-hull statement additionally requires equal durations and integer H/Δ. It is a statement about feasible actions, not convex-objective minima: b=P=(2,2), one allowed hour and S=[1−x1]+ +[1−x2]+ give M1 minimum1 and relaxed minimum0.

Inclusions: b≤P implies x/P≤x/b≤u, so F_M1⊆F_M1LP⊆F_M2. Hybrid α tightens debit; interpolation β loosens it. Primary minima reverse these inclusions, and exact lexicographic pairs are ordered, while secondary energy alone need not be. Strict inclusion is not automatic; full-request b=P can make the continuous domains equal.

Protection: fixed q≥1 gives Dq≥D2 and F_Cq⊆F_M2. Uniform qbar is exactly physical allowance H/qbar without new floor-rounding. Mean-one plus q≥1 and positive W forces every q=1; the positive mean premium1.25 is therefore substantive. A convex g with g(0)=0, g(1)=1 lies below its chord z; a nontrivial conservative convex surcharge cannot preserve both endpoints.

QP: W>0 and λ>0 give a strictly convex objective on a compact feasible domain containing qbar·1 when0≤ξ≤γ. Interior mean premium and nonconstant ordered score means admit direction(1/W1,0,−1/W3) with negative linear derivative at uniform; hence differentiation follows. The executed W, thresholds and means reconstruct exactly; mean residual1.55×10^-11 FDE hours and free KKT difference below10^-12 support the given unique card.

Dual bound: add λ(Dx−H) and ζ(a x+β−θ) to the minimization objective. Nonnegative multipliers and Σζ≤1 make the θ infimum finite. Box minimization gives Σ b min(r,0), which is mandatory. Counterexample: θ≥1−x,0≤x≤1,θ≥0 has optimum0. ζ=1 yields β=1, r=−1, hence L=1−1=0; omitting the box term falsely gives1. Nonzero r can reflect active box bounds as well as arithmetic error.

Screening: a hinge whose full-box affine maximum is≤0 is inactive. Whole-coordinate zeroing preserves value when every potentially positive hinge has nonnegative coefficient and debit cannot increase; reset the M1 support to zero. This does not justify arbitrary positive M1 contraction below its materiality floor. Variation debit is a counterexample to generic monotonicity: with ηh0=2, unit durations, z=(1,1,1) costs5 but(1,0,1) costs6. The sufficient ηh0≤minΔ condition restores monotonicity. No optimized variation/progressive family was introduced.

Chronology: nonnegative debit and each window budget imply nonnegative next balance after only the committed block; telescoping gives annual feasibility in exact arithmetic. The numerical code clips tiny negative balances. The fresh ledger report records those residuals and distinct guarded annual witnesses instead of declaring raw trajectories exactly feasible.

Exposure witness: nonnegative positionwise contraction preserves the box and C debit; E_EC≤E_U implies qbar E_EC/P≤qbar E_U/P≤H. Soft thermal slack preserves algebraic feasibility. Thus each improved EC action is also uniform-feasible. Convexity even supplies an annual improving segment from the uniform witness; no prospective controller result follows.

The actuator floor is the minimum signed-row exceedance over the full independent box. It lower-bounds achievable row exceedance, not relief; the companion caption was corrected. Temporal budgets can only restrict that box. Different rows may demand conflicting actions, so a summed floor is valid but need not be attainable.

5. Numerical reconciliation and immutable historical outcomes

The fresh read-only replay independently re-evaluates all 41 annual/raw/guarded/retrospective objects through complete8837-row chronology and 14 epoch ledgers. It compares the underlying arrays with original hashes and recovers committed actions/activations, solver routes, prospective balance checks and final balances. All124 numerical table cells pass explicit display-to-source arithmetic. Recorded runtimes/memory remain documentary measurements, not fresh performance benchmarks.

FINAL_TABLE_VERIFICATION.csv maps every table cell to exact evidence and script. FINAL_NUMERICAL_RECONCILIATION.json recomputes each outcome denominator, energy/FDE difference, matched ceiling, actuator row and fourth-quarter sum. Fresh fitting/evaluation derivative arrays reproduce the

frozen card; all 6,993,216 nomination values and 114 capacities match the deterministic formula. Stored AC bus/branch/loading rows reaggregate to the recorded summaries.

Changed numerical displays are explicitly classified: (i)new retained lower bounds; (ii)outward lower/upper endpoint rounding; (iii)previously corrected2029 rolling M2=100.252003 million retained; (iv)GiB labels retained for1024^3 measurements. No historical trajectory, solver output, allowance, parameter or comparator was changed. Full-precision records, rather than rounded cells, determine strict separation.

Original rolling M2 improves S versus M1 by22.1–26.8% with38.4–40.6% more FDE and 4.353,5.490,6.905 extra TWh. The M1-energy-matched retrospective comparison is0.199% worse,0.261% worse and 0.163% better. All342 matched ceilings pass. The proper conclusion is no consistent equal-position-energy rolling advantage, not a general annual domain disadvantage.

6. Result-to-script traceability and computational routes

Historical physical construction: jobs/input_build.py and tpwrs/network.py. Historical orchestration: jobs/primary.py. Continuous M1-LP/M2/positive-weight rules: tpwrs/annual.py. Generic guarded mixed-integer M1: tpwrs/model.py. Chronological commitment: tpwrs/rolling.py. Original witnesses: jobs/certificates.py. Conservative fitting/campaign: jobs/debit_rule_family.py and tpwrs/debit_rules.py. Exposure controls: the separately archived diagnostic source, not a new optimization in this task.

Algorithm 1 was corrected to the source: initial tight J_t(0) plus validated selector hints; StageI aggregate/maximum support-gap closure and complete replay; StageII accepts an actual replay on the primary cap, so every hourly epigraph need not be tight. The M1 route is explicitly separate. Toleranced StageII is not falsely called the exact lexicographic optimum.

New Gate 1 code is standalone and never imports a protected project module. Exclusive start markers prevent repeats. The pre-optimization compressed-MPS export incident used zero optimize calls; its logs remain. Plain MPS plus Python gzip fixed only I/O. All three actual optimize calls are separately marked and logged. GATE1_VERIFICATION_CODE includes a no-argument, read-only PyCharm inspector that recomputes saved finite-box arithmetic without a solver.

FINAL_SOURCE_TO_RESULT_QUICK_MAP.csv and the 11-page updated laptop guide give exact Windows locations, data/model/script maps, read-only versus optimizing commands, RUN_00/Stage05C integrity pitfalls, and safe recovery of a separate review copy. Large matrices, full duals and private solver logs remain outside the publisher source ZIP.

7. Physical-model assessment and scope of the annual certificate

The frozen evaluator has 45 operational buses,114 directional positions and 94 retained states. Its95 elements and 93 non-islanding outages yield95+93·94=8837 monitored rows/hour. Signed restriction/counter-injection columns and affine transformer effects are preserved. One x is tested in every state; no corrective recourse, unit commitment, storage dynamics, ramping or feasible generator-headroom constraints are optimized.

The383-bus structural layer has 730 lines/33 transformers; the 45-bus operational layer92 lines/3 transformers. Multibus admission allows relative error cutoffs2.135 and 5.212, alongside sign/ranking filters. These are broad empirical thresholds, not critical-branch absolute-error bounds or proof of electrical equivalence. The frozen balancing system is neither a proven inner nor outer approximation of realistic redispatch.

The3 annual M2 peaks occur on line45 during outage44. At their exact row-hours the 2501.081366-MW rating and independent full-box extrema leave excess floors1566.930,1208.853 and 800.339 MW. This identifies an actuator-limited component within fixed balancing, not the entirety of annual residual or a proof that reinforcement is necessary.

Fresh AC aggregation confirms2106 records over 18 hours,3 states/hour,13 variants and 3 power factors. All converge, but846 pass thermal limits. Per162 named cases, background passes162, requests0, rolling M1=48, rolling M2=84. The292.45% maximum is registry-wide. The324 normalized-index comparisons contain270 DC ties; all 54 non-ties retain ordering. No premium AC outcomes or independent complete device-level Q-vector reconstruction are claimed.

These limitations are material referee risks for operational recommendations. They do not invalidate an explicitly conditional comparison within the same fixed numerical DC testbed. The manuscript preserves them rather than moving a contradictory security claim to the companion.

8. Premium mechanism, novelty and exposure interpretation

The new construction is structurally distinct from old M3: it scores the primary requested-state right derivative over all signed rows, whereas the old μ·FDE labels are StageII secondary-energy sensitivities at fixed primary cap. It uses conservative q≥1 and a positive pooled mean1.25 instead of the old discount-capable mean-one normalization. Neither generic cutting planes nor duality is a new invention.

Frozen low-to-high relief rates are1.4073554014,1.3426166214,1. The uniform control is1.25. High-relief rate-one hours can use up to H physical FDE, while uniform caps every position at0.8H; expensive buckets charge more than uniform. Actual box-availability tests prove that the two premium domains are nonnested.

Verified2029 guarded outcomes: C residual103.924090 million versus U120.434609 million, a13.709% improvement; ordinary rolling M2 is better at100.252003 million. C energy13.766142 TWh exceeds U11.771910 TWh by1.994232 TWh; FDE67501.715 versus66734.850 is about 1.15% greater. Raw fourth-quarter energy is0.877986 versus0.227597 TWh. These stored facts are consistent with changed timing and position composition, but do not prove a causal reserve-for-Q4 mechanism.

The2029–2031 C-versus-U gains are13.71%, 15.94%, 15.92%; EC gains are8.21%, 9.14%, 9.76% with342 exact participant ceilings and both premium domains satisfied. Thus improved chronological allocations exist relative to the saved uniform output under position-specific energy/FDE ceilings, but they are also uniform-feasible and retrospective. There is no tested prospective matched-exposure controller or proof that differentiated debit is necessary.

Main-paper inclusion is justified as a transparent secondary construction and controlled comparison, with adverse controls. The evidence does not justify presenting it as a superior contract, an equal-protection efficiency theorem, or a solution outperforming ordinary M2.

9. Data provenance and information audit

All24 registered immediate input files were freshly SHA-256 checked against the frozen registry. The parent physical array hash and all annual case/time/network identities match. The deterministic nomination formula reproduces6993216 values and 114 capacities exactly; this establishes formula identity, not the precise historical executed generator revision.

Provincial scenario profiles cover12 provinces and 96408 UTC hours in 2025–2035; the study uses61344 hours in 2025–2031. Component mappings are nonnegative and column-normalized. Future weather shapes are reused from 2017/2016. These are constructed scenarios and technology/day/hour requests, not observations of future customer behavior.

Requested injections use v, not clipped b; eligibility constrains restriction actuators. The audit reproduces this declared input semantics without silently zeroing requests or changing masks. Some historical public download revisions, exact network source commit and processing-generator bindings remain unavailable. Immediate file identity does not establish end-to-end source regeneration.

The record binds one fitting-designed candidate,2028 accept/reject validation, then2029–2030 evaluation and 2031 confirmation. Both premium source hashes match the freeze; all 8 experiment identities use the same card, and 3997296 evaluation assignments reconstruct exactly. The freeze and code show the recorded sequence. They do not prove that the overall research-design process was untouched by earlier duration/depth outcomes.

The controller uses realized deterministic48-hour profiles and requested-state scores, with24-hour commits and no continuation value. A prospective version requires timestamped forecasts and legally timed frozen announcements. The hourly study does not implement07:00 clock-quarter TDTR administration.

10. Tables, figures and visual audit

The final main contains4 numerical tables, debit geometry Figure 1, conservative rate-card Figure 2, and a separate Algorithm 1. The old AC Figure 3 is absent. AC evidence is a numerical paragraph with explicit policy coverage. The companion retains2 tables and the original architecture PDF, correctly labeled as schematic rather than verified geography/topology layout.

Every numerical table cell is independently mapped and re-evaluated. The geometry vertices and M1 materiality segments agree with their two-hour definitions; the rate plot uses frozen coefficients and reference1.25. The architecture generator and exact edge-to-source layout provenance remain unrecovered. Its bytes are identical to the authoritative latest ZIP, not a newly invented network drawing.

Editorial shortening removed repeated prose while preserving all material assumptions, proofs, adverse controls and engineering qualifications. The initial page review detected an accidentally removed adjacent companion float; it was restored unchanged and the final source/ref check passed. An inline forecast formula was displayed to remove an overrun; the actuator caption was corrected from relief to exceedance.

The final source directories were rebuilt cleanly without cached auxiliary files. Main/report are10/7 pages. Fonts/margins are unchanged from the supplied10-point IEEEtran main and 11-point22-mm-margin report. Optional microtype was omitted after a local package/kernel incompatibility; no font-size, margin or spacing reduction was used to force page counts. Unmodified local xurl and IEEEtran style dependencies are included for reproducibility. Every final rendered page is reviewed before sealing.

11. Bibliography and scientific positioning

Seventeen pre-existing DOI entries were resolved against primary Crossref metadata; transient rate-limit responses were retried and retained. Original publisher metadata supplies Chao/Wilson DOI 10.1287/opre.38.4.598. The pandapower entry now includes volume 33, issue 6, pages 6510–6521. All cited keys compile. Full metadata and correction records are retained.

The primary TU Delft dataset record identifies Wouter Zomerdijk, GitHub,15 April 2025, and DOI 10.5281/zenodo.15223447. This is distinct from the 2022 research article and is not claimed to identify the exact historical local revision. The dataset DOI is never used as a technical-report DOI.

The related arXiv2610.15510 record verifies Kryeziu, Zhu, Ursavas and 11 October 2026 submission. Its topic is product/calendar evidence, admission/routing and interventions; PaperB holds the portfolio fixed and studies depletion under partial restrictions. The bibliography and introduction make that distinction without claiming annual interruption budgets, cutting planes or standard duality as new inventions.

The introduction now distinguishes annual discretionary interruption service from priority-service design and monthly constrained load control. The claimed contribution is the alternative partial-action debit mapping, geometry and network consequences, with a conditional computational witness. The separately prepared technical report is cited normally as a University of Groningen technical report, version 1.0; no fictional Zenodo DOI or accepted-supplement status is inserted.

12. IEEE formatting and submission compliance

Primary IEEE PES guidance checked17 October 2026 specifies10 initial pages, a150–200-word abstract and up to10 keywords. The final main has 10 IEEEtran pages, a167-whitespace-word self-contained abstract without citations or unexplained abbreviations, and 6 keywords. Standard mathematical/theorem numbering and all cross-references/citations pass the clean build.

Current portal: https://ieee.atyponrex.com/journal/tpwrs-pes . Primary preparation/author-information URLs and their verified scopes are in EXTERNAL_PRIMARY_REFERENCES.csv. The main retains the full scientific argument; the companion is independently prepared, not automatically accepted IEEE supplementary material.

Acknowledgments identify OpenAI ChatGPT/Codex, affected sections and the level of drafting, code and numerical assistance, with a system citation. They assign author responsibility without asserting undocumented author verification or approval. Authors must confirm this disclosure matches actual use before submission.

The Overleaf ZIP is main-compilation sources only. The larger submission-lock ZIP is an author handoff with the requested reports and preparation documents; it is not an instruction to upload every file to IEEE. Private solver logs, models and dual arrays are outside both publisher source uploads. No IEEE submission or external message was sent.

13. Zenodo technical-report readiness

The standalone7-page report includes the author-supplied author order, common affiliation/contact, abstract,17 October 2026 date, version 1.0 pre-deposit status, definitions/proofs, new numerical qualification, tables, original schematic and references. Its relation to the main manuscript is explicit. The source tree and all required local assets compile independently.

ZENODO_METADATA.md, ZENODO_DESCRIPTION.md, ZENODO_CITATION_PENDING.md and ZENODO_RELEASE_CHECKLIST.md are prepared locally. No report DOI has been verified. No DOI was fabricated or reserved, no record modified, and no file uploaded/published. No report license was selected for the authors.

Public citation is a real remaining release condition. The human authors must authorize deposit, approve licensing and metadata, verify the assigned DOI, and replace the pre-deposit citation. That metadata change requires

recompile, page/reference review and new hashes in a new versioned lock. The current package remains a truthful pre-deposit sign-off.

14. Remaining limitations and adversarial referee issues

Issue / location	Objection and evidence	Severity / disposition
Annual certificate; main §V,VII	Original duals absent; new full-box solves retain complete vectors and reproduce strict margins. Full interval enclosure remains absent.	Major evidence gap addressed by new evidence; precision qualification remains.
Algorithm1; main §V	Empty initial support and identical StageI/II closure would misdescribe frozen code. Source initializes J(0) and uses StageII replay cap.	Material exposition corrected; no code/result changed.
Premium claim; main §VII-D,E	More energy/FDE than uniform; ordinary M2 better. Improved EC witnesses are also uniform-feasible.	Major interpretation risk controlled; no superior-contract claim.
Physical model; main §II,VI,VII-F	Broad aggregation errors,fixed balancing without headroom,soft thermal and selected AC failures.	Major operational-validity limit retained; conclusions scoped to testbed.
Inputs; main §VI	Deterministic nominations and incomplete upstream public/generator revision binding.24 current hashes and formula replay do not close provenance.	Reproducibility limitation remains explicit.
Companion Figure1	Original asset supplied; exact layout/edge generator not recovered.	Schematic only; no geographic or independently reconstructed topology claim.
Companion hull and floor wording	Convex-hull equality does not imply equal convex minima; floor bounds exceedance rather than relief.	Minor mathematical exposition corrected with proofs retained.
Submission/Zenodo	Report DOI,license,coauthor approval and metadata confirmation outstanding.	Concrete human release conditions; conditional handoff only.

No heavy physical experiment or new family was launched to resolve these limits. They are not hidden in the report while a contradictory main claim remains. There is no defensible invented journal acceptance probability.

15. Scientific recommendation and decision boundary

CONDITIONAL GO is supported for the final scoped manuscript: exact action-set results, independently retained guarded annual separations, honest adverse controls and explicit model limits. The new score is a distinct methodological construction beyond old M3, while its empirical role is secondary. It improves the saved uniform-premium chronological allocation but not ordinary M2; participant-specific controls support better retrospective allocations, not differentiated-contract necessity or prospective superiority.

The authors should retain the accounting-dependent performance framing. A claim of realistic Dutch secure dispatch, regulatory equivalence, equal customer protection or a new global contract-performance theorem would exceed the evidence and change this decision. Stronger interval language requires separately completed outward-rounded work; it is not silently inferred from the large margins.

Proceed to human review only after the final protected-source comparison and package checks pass. Remaining author/DOI/license steps are specified rather than presumed complete.

Final preservation check: 713,752 protected files (655,496,382,007 bytes) match their complete before-work SHA-256 inventory; no changed, missing or added file. Inventories are sequential, not atomic snapshots. All directly accessed reconstruction inputs also have specific retained source hashes.

16. Coauthor approval and final release checklist

Pending human approval: all four authors confirm authorship/order, affiliations, email, originality/related work, mathematical statements, numerical-object labels, AI contribution disclosure, and the testbed/engineering scope. No coauthor approval has been obtained or inferred by this task.

Pending human release: choose/authorize the technical-report license; separately authorize Zenodo deposit; verify its actual DOI; update the definitive scholarly citation; clean-compile and visually recheck; create a new versioned source/PDF/ZIP hash lock. Then authorize IEEE submission separately.

Completed computational checks are indexed in the Gate 1 report,124-cell table audit,41-object replay,14-ledger record,24-input hash check, derivative/card verification, clean-build status and final integrity report. The publication package excludes large private evidence but points to its exact local locations.

The frozen repository, prior audit and original attachments/ZIP remain the scientific reference records. All corrected prose, independent certificates and package artifacts are separate. The authors retain responsibility for final scientific and submission decisions.